



Chapter # 08

Thermoregulation Homeostasis



Student Learning Outcomes

At the end of this chapter, students will be able to:

- [B-12-R-74] Define thermoregulation and explain its needs.
- [B-12-R-75] Classify animals on the basis of the source of body heat i.e., ectotherms and endotherms.
- [B-12-R-76] Classify the animals on the basis of the ability to thermoregulate i.e., poikilotherms and homeotherms.
- [B-12-R-77] Describe the regulatory strategies in man for thermoregulation.
- [B-12-R-78] Describe three elements i.e., receptors, control center and effectors which operate homeostatic mechanisms.
- [B-12-R-79] Relate the homeostatic mechanisms with the negative and positive feedback systems.
- [B-12-R-80] Differentiate between osmoconformers and osmoregulators.
- [B-12-R-81] Define osmoregulation.
- [B-12-R-82] Explain the problems faced by osmoregulators.
- [B-12-R-83] Explain the different methods of osmoregulation found in freshwater, marine water and terrestrial.

Living organisms must maintain a stable internal environment to survive, even when external conditions change. This ability is called **homeostasis**, and one of its most important aspects is the regulation of body temperature, known as **thermoregulation**.

Temperature strongly influences the speed of biochemical reactions, the structure of proteins, and the functioning of enzymes. If body temperature rises too high (**hyperthermia**), enzymes may denature and vital processes can collapse. If temperature falls too low (**hypothermia**), reaction rates slow down and cells cannot function properly. Thus, keeping body temperature within an optimal range is essential for life.

Different animals have evolved various strategies to cope with temperature fluctuations. Some, like fishes, reptiles, and amphibians, rely largely on external heat sources and are called **ectotherms**. Others, such as birds and mammals, generate and regulate their own body heat, and are termed **endotherms**.

In humans, thermoregulation is controlled by the **hypothalamus**, which acts as a **thermostat**. It receives information from temperature receptors, compares it with the set point (around 37 °C), and activates responses through effectors such as sweat glands, blood vessels, and muscles. These responses may involve **heat-loss mechanisms** (like sweating and vasodilation) or **heat-gain mechanisms** (like shivering and vasoconstriction).

Understanding thermoregulation not only explains how organisms maintain balance with their environment but also highlights the intricate coordination of **receptors, control centers, and effectors** that sustain life.

SLO [B-12-R-74]

Define thermoregulation and explain its needs.

THERMOREGULATION

Thermoregulation is the biological process by which organisms maintain a stable internal body temperature within a narrow and optimal range, despite fluctuations in the external environment. It involves physiological, behavioral, and structural adaptations that prevent harmful variations in body temperature. In humans and other warm-blooded animals, thermoregulation is primarily controlled by the hypothalamus, which acts as the body's "thermostat."



Importance and Needs of Thermoregulation

Maintaining a constant internal temperature is essential for survival because most biochemical processes and enzymes in living organisms are temperature-sensitive. Even small deviations from the normal range (36.5–37.5°C in humans) can disrupt homeostasis and impair body functions. The main needs of thermoregulation are:

1. Enzyme Function and Metabolism

Enzymes are highly specialized proteins that act as catalysts for almost every biochemical reaction occurring in living cells. Their activity is strongly dependent on temperature, as they function efficiently only within a narrow optimal range. In humans, this optimal temperature is around 37 °C. If body temperature rises beyond this range, a condition called hyperthermia occurs. Under hyperthermia, enzymes begin to lose their three-dimensional shape, a process known as denaturation, which severely reduces their ability to catalyze reactions. As a result, essential processes such as respiration, digestion, and protein synthesis slow down or may even cease entirely. On the other hand, if the body's temperature falls below the normal range, hypothermia sets in. At low temperatures, enzymes do not denature but their activity becomes sluggish, leading to a sharp decline in metabolic rate. This reduced metabolic efficiency compromises the production of energy and weakens cellular functions. Thus, thermoregulation is vital for maintaining the precise temperature conditions necessary for enzyme activity and overall metabolism.

2. Cell and Organ Protection

Temperature extremes pose a direct threat to the integrity of cellular structures and organs. At high temperatures, proteins and lipids that form the building blocks of cells undergo structural damage. Proteins may unfold or clump together, disrupting normal cellular machinery, while lipids in membranes become excessively fluid, weakening the barrier that protects cells. Nucleic acids such as DNA and RNA are also sensitive to high heat, which can cause mutations or breakage of their molecular bonds. Conversely, at very low temperatures, cells face the risk of ice crystal formation within their cytoplasm. These ice crystals are sharp and can puncture delicate membranes and organelles, leading to irreversible damage and cell death. Vital organs such as the heart, liver, and kidneys are especially at risk in such conditions, since their failure can quickly threaten survival. Thermoregulation, by keeping body temperature within safe limits, provides essential protection against these potentially lethal effects.

3. Homeostasis of Vital Processes

The maintenance of a constant body temperature is fundamental to the smooth functioning of vital physiological processes. For example, respiration depends on temperature-sensitive enzymes in mitochondria to generate ATP efficiently. Circulation is also influenced, as both high and low temperatures alter the viscosity of blood and the efficiency of the heart in pumping it. Nerve conduction, which relies on precise ion movements across membranes, becomes slowed or erratic when temperatures deviate from the normal range. Organs such as the brain and the heart are especially sensitive to temperature disturbances. The brain requires a steady environment to maintain consciousness, coordination, and memory, while the heart depends on precise biochemical signals for rhythmic contractions. Even a slight increase above normal body temperature can lead to confusion, seizures, or cardiac stress, while a drop below the normal range can result in loss of coordination or unconsciousness. Therefore, thermoregulation is indispensable in sustaining the internal environment required for these vital processes.

4. Survival in Fluctuating Environments

Living organisms are constantly exposed to environmental changes, such as day-to-night temperature variations, seasonal shifts between summer and winter, and geographic differences between hot deserts and cold polar regions. Without thermoregulation, organisms would not be able to adapt to these fluctuations and would risk death during unfavorable conditions. For instance, desert animals must prevent overheating during the day and avoid chilling during the night, while polar animals need insulation and behavioral mechanisms to withstand freezing conditions. Thermoregulation enables animals to remain active, find food, reproduce, and escape predators in both hot and cold habitats. In humans, it allows us to live in and adapt to diverse environments across the globe—from the scorching tropics to icy mountain ranges—by maintaining internal temperature stability despite extreme external variations. Thus, thermoregulation ensures survival and functionality in ever-changing climates.

5. Energy Balance

Thermoregulation also plays a crucial role in managing the body's energy resources. Maintaining body temperature within a stable range minimizes unnecessary energy expenditure, allowing organisms to direct energy toward growth, reproduction, and activity rather than continuous adjustment to environmental



changes. For example, in the absence of thermoregulation, the body would need to accelerate metabolism drastically during cold conditions to generate heat or slow it excessively during hot conditions to prevent overheating. Such frequent and extreme metabolic changes would consume vast amounts of energy, leaving little for other essential processes. By keeping temperature stable, thermoregulation prevents wastage of energy and ensures efficient utilization of nutrients and stored reserves. This balance between temperature regulation and energy conservation is especially critical for survival during times of food scarcity, migration, or hibernation.

Thermoregulation is the process of maintaining a constant internal body temperature to ensure normal enzyme activity, protect cells, maintain homeostasis, and support survival in changing environments. It is an essential aspect of homeostasis and vital for the continuity of life processes.

SLO [B-12-R-76]

Classify the animals on the basis of the ability to thermoregulate i.e., poikilotherms and homeotherms.

CLASSIFICATION OF ANIMALS ON THE BASIS OF THERMOREGULATION

Animals differ in how they regulate their internal body temperature when exposed to environmental changes. Based on this ability, they are classified into two broad categories: **poikilotherms** and **homeotherms**. This classification highlights the distinct strategies animals have evolved to cope with the thermal challenges of their habitats.

1. Poikilotherms (Cold-blooded Animals)

Poikilotherms are animals whose internal body temperature fluctuates with the temperature of their external environment. They do not possess efficient physiological mechanisms to maintain a constant internal temperature. Instead, their metabolic activities and body functions rise or fall according to ambient conditions. For example, when exposed to heat, poikilotherms become more active due to an increase in enzymatic activity, whereas in colder environments, their activity slows down drastically.

The lack of internal regulation makes poikilotherms heavily dependent on environmental conditions for survival. To cope, they rely on **behavioral adaptations**, such as basking in the sun to raise body temperature or seeking shade, burrows, or water to avoid overheating. Many reptiles, amphibians, fishes, and invertebrates are poikilothermic. For instance, lizards are often seen basking on rocks in the morning to absorb heat, and frogs become sluggish during winter as their body processes slow down.

While poikilothermy conserves energy—because these animals do not burn large amounts of food for heat production—it also restricts them. They cannot remain active in extreme cold or heat for long periods, and their distribution is limited to environments where temperature fluctuations are within tolerable limits.

2. Homeotherms (Warm-blooded Animals)

Homeotherms are animals that maintain a relatively constant internal body temperature, regardless of external environmental conditions. This is achieved through well-developed **physiological mechanisms**, such as sweating, panting, shivering, vasodilation, vasoconstriction, and changes in metabolic rate. The hypothalamus acts as the central control system, ensuring that the body's temperature remains within a narrow range (for humans, typically around 36.5–37.5 °C).

Because homeotherms regulate their own temperature, they can remain active across a wide range of environmental conditions, from deserts to polar regions. Birds and mammals are classic examples of homeotherms. For instance, polar bears maintain body warmth through thick fur and fat insulation even in freezing conditions, while camels regulate heat storage and water balance to survive in hot deserts.

However, maintaining homeothermy requires a **high metabolic rate**, which in turn demands a continuous supply of food and oxygen. This makes homeotherms energetically more expensive than poikilotherms. Despite this cost, the advantage lies in their ability to survive, reproduce, and remain active in diverse and often harsh climates.

In conclusion, animals can be classified into two groups based on thermoregulation: **poikilotherms**, which allow their body temperature to fluctuate with the environment, and **homeotherms**, which maintain a stable internal temperature through physiological regulation. Poikilothermy conserves energy but restricts activity in extreme climates, while homeothermy allows independence from environmental conditions but requires greater energy expenditure. Both strategies illustrate the diversity of adaptations that animals have evolved to survive in their respective habitats.



CLASSIFICATION OF ANIMALS ON THE BASIS OF THE SOURCE OF BODY HEAT

Animals can also be classified according to the primary source from which they obtain body heat. This classification distinguishes between organisms that rely on the external environment for heat and those that generate it internally through metabolic processes. On this basis, animals are grouped into two main categories: **ectotherms** and **endotherms**.

1. Ectotherms

Ectotherms are animals that depend primarily on heat gained from their external environment to regulate body temperature. Their internal heat production through metabolism is relatively low and insufficient to maintain a stable temperature on its own. As a result, ectothermic animals exhibit wide fluctuations in body temperature that closely mirror environmental changes.

To survive, ectotherms adopt **behavioral strategies** to regulate heat. For example, reptiles like snakes and lizards bask in the sun to absorb heat and raise their body temperature for activity, while in excessively hot conditions they retreat to burrows, shaded areas, or water bodies to avoid overheating. Amphibians, fishes, and most invertebrates are also ectothermic, relying on the warmth of the sun or surrounding water to function efficiently.

The advantage of ectothermy is that it conserves energy, as these animals do not burn large amounts of food to generate heat internally. This allows them to survive on smaller food supplies compared to endotherms. However, the disadvantage is their dependence on environmental temperatures. In cold conditions, ectotherms become sluggish and unable to maintain high levels of activity, which limits their survival in extreme or fluctuating climates.

2. Endotherms

Endotherms are animals that generate most of their body heat through internal metabolic processes, such as cellular respiration. This internally produced heat allows them to maintain a relatively constant body temperature, independent of the external environment. Endotherms use both **physiological and structural adaptations** to balance heat production and heat loss. Examples include shivering to produce heat, sweating or panting to cool down, insulating fur or feathers, and fat layers such as blubber in whales and seals.

Birds and mammals are typical endotherms. A sparrow, for example, maintains high body temperature to sustain flight even in cool mornings, while polar bears and arctic foxes use thick fur and fat insulation to survive freezing polar regions. Endothermy provides a significant advantage by enabling activity in a wide range of habitats and climates, as well as during both day and night.

The cost of endothermy, however, is **high energy demand**. Since internal heat production requires rapid metabolism, endotherms must consume large amounts of food and oxygen. This constant energy requirement makes them more vulnerable to food scarcity, but it is balanced by the benefit of environmental independence and continuous activity.

Animals are classified on the basis of their heat source into **ectotherms**, which rely on environmental heat and show fluctuating body temperatures, and **endotherms**, which generate their own heat metabolically and maintain stable body temperatures. Ectotherms conserve energy but are dependent on external conditions, while endotherms remain independent of the environment but require large amounts of energy. This classification highlights the contrasting strategies animals use to meet the universal challenge of temperature regulation.

REGULATORY STRATEGIES IN MAN FOR THERMOREGULATION

Humans, being homeothermic and endothermic animals, maintain their internal body temperature within a narrow range of about $36.5\text{--}37.5\text{ }^{\circ}\text{C}$, regardless of environmental fluctuations. This regulation is controlled mainly

by the **hypothalamus**, which functions as the body's thermostat. The hypothalamus constantly monitors blood temperature and, when deviations are detected, initiates physiological and behavioral responses to restore balance. These regulatory strategies can be divided into two categories: responses to overheating (**heat-loss mechanisms**) and responses to excessive cooling (**heat-gain mechanisms**).

1. Heat-Loss Mechanisms (Regulation in Hot Conditions)

When the external temperature is high, or when internal heat is generated through strenuous muscular activity, the human body must prevent overheating to maintain homeostasis. Overheating can be dangerous, as it disrupts enzymatic functions, accelerates dehydration, and may lead to heat stroke. To avoid these outcomes, the hypothalamus activates several coordinated mechanisms that promote heat loss and restore body temperature to normal.

- **Sweating (Evaporative Cooling):** Sweating is the most effective cooling mechanism in humans. Sweat glands located in the dermis release fluid onto the surface of the skin. As this sweat evaporates, it absorbs a large amount of latent heat from the body, thereby cooling the blood flowing beneath the skin. This process works best in hot and dry environments where evaporation is rapid. However, in humid conditions, evaporation is slowed down, which makes sweating less efficient and explains the discomfort experienced in such climates.
- **Vasodilation of Skin Arterioles:** Another key response is vasodilation, in which blood vessels near the skin surface widen. This increases the flow of warm blood close to the surface, where heat can escape into the environment through **radiation, convection, and conduction**. The skin often becomes flushed or reddish as a result of increased blood flow. By channeling heat outward, vasodilation prevents dangerous rises in core body temperature.
- **Reduced Metabolic Rate:** The body also regulates its internal heat production by lowering the metabolic rate. By slowing down chemical reactions and energy release, the production of extra heat within cells is minimized. This conservation of energy reduces the risk of further overheating and supports the body in maintaining balance.
- **Behavioral Adjustments:** In addition to physiological mechanisms, humans consciously adopt behavioral strategies to regulate temperature. Examples include wearing light and loose-fitting clothing, drinking water to replace fluids lost through sweating, reducing strenuous activity, and seeking cooler environments such as shade or air-conditioned spaces. These behaviors complement internal regulatory mechanisms and play a crucial role in preventing dehydration and heat-related illnesses.

Through the combined actions of sweating, vasodilation, reduced metabolism, and behavioral strategies, the body maintains a safe internal temperature during hot conditions. This integrated system ensures the protection of enzymes, cells, and vital organs from the harmful effects of hyperthermia, thereby preserving normal physiological function even under extreme heat.

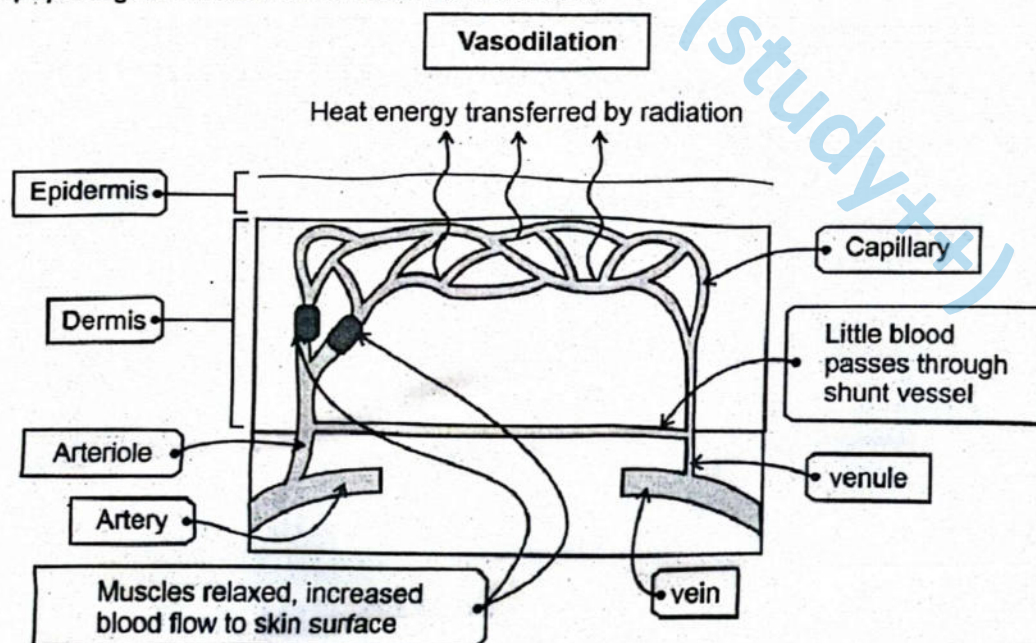


Figure: Mechanisms of vasodilation

2. Heat-Gain Mechanisms (Regulation in Cold Conditions)

In cold environments, the body must conserve existing heat and produce additional warmth to prevent hypothermia. A fall in body temperature slows down enzyme activity, reduces metabolic efficiency, and threatens vital organ function. To counter these effects, the hypothalamus activates a series of **physiological** and **behavioral** mechanisms designed to retain or generate heat.

- **Shivering:** One of the most immediate responses to cold is **shivering**, which consists of rapid, involuntary contractions of skeletal muscles. These contractions require energy and therefore increase the rate of cellular respiration, which in turn produces heat as a by-product. Although shivering can be physically tiring, it is highly effective in raising body temperature during sudden exposure to cold.
- **Vasoconstriction of Skin Arterioles:** Another important strategy is **vasoconstriction**, in which arterioles supplying blood to the skin narrow. This reduces the flow of warm blood to the surface, thereby minimizing heat loss through radiation, convection, and conduction. As a result, the skin may appear pale or bluish, particularly in the fingers, toes, and lips. By preserving heat in the body's core, vasoconstriction helps protect vital organs from dangerously low temperatures.
- **Piloerection (Goosebumps):** The contraction of tiny muscles at the base of hair follicles causes hairs to stand upright, a response known as **piloerection** or "goosebumps." In ancestral mammals with dense fur, this mechanism trapped an insulating layer of air close to the skin, reducing heat loss. In humans, where body hair is sparse, this response is largely ineffective for insulation but remains as a vestigial mechanism.
- **Increased Metabolic Rate:** Cold exposure also stimulates hormonal responses that increase heat production. The **thyroid gland** secretes thyroxine, which raises the basal metabolic rate, while the **adrenal medulla** releases adrenaline, which accelerates metabolism and enhances heat generation. These hormonal adjustments ensure a sustained supply of energy and warmth, especially during prolonged exposure to cold conditions.
- **Behavioral Adjustments:** In addition to automatic physiological responses, humans consciously adopt behaviors that help conserve and generate heat. These include wearing insulating clothing, covering exposed body parts, seeking shelter from wind and cold, consuming warm food and drinks, and engaging in physical activities such as exercise to stimulate muscle activity and heat production.

Through the combined action of shivering, vasoconstriction, piloerection, metabolic adjustments, and behavioral strategies, the human body maintains core temperature within the normal range. These mechanisms are critical for keeping the brain, heart, and other essential organs functional, ensuring survival in cold environments and preventing the life-threatening effects of hypothermia.

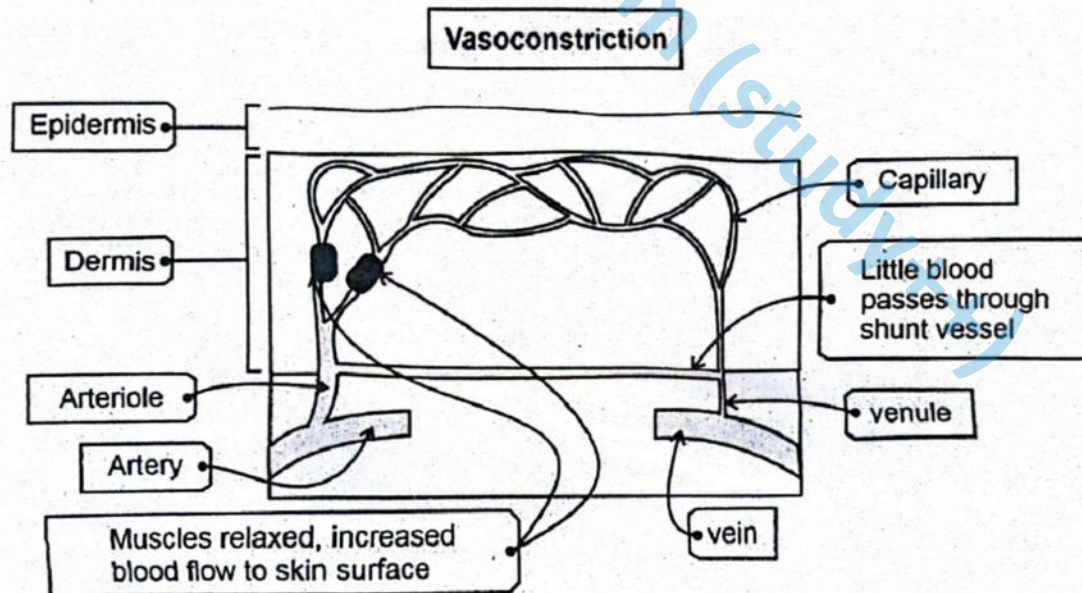


Figure: Mechanisms of vasoconstriction

3. Role of the Hypothalamus

The hypothalamus integrates all information related to body temperature. Thermoreceptors in the skin detect changes in external temperature, while internal thermoreceptors monitor blood temperature. The

hypothalamus compares this information to the body's set point and triggers appropriate corrective responses. This constant regulation exemplifies **homeostasis**, ensuring survival in diverse climates. In humans, thermoregulation is achieved through a combination of physiological and behavioral strategies controlled by the hypothalamus. In hot conditions, sweating, vasodilation, and reduced metabolism promote heat loss, while in cold conditions, shivering, vasoconstriction, and increased metabolism promote heat conservation and heat gain. These mechanisms, along with conscious behavioral actions, maintain body temperature within the narrow optimal range necessary for normal enzyme function and overall survival.

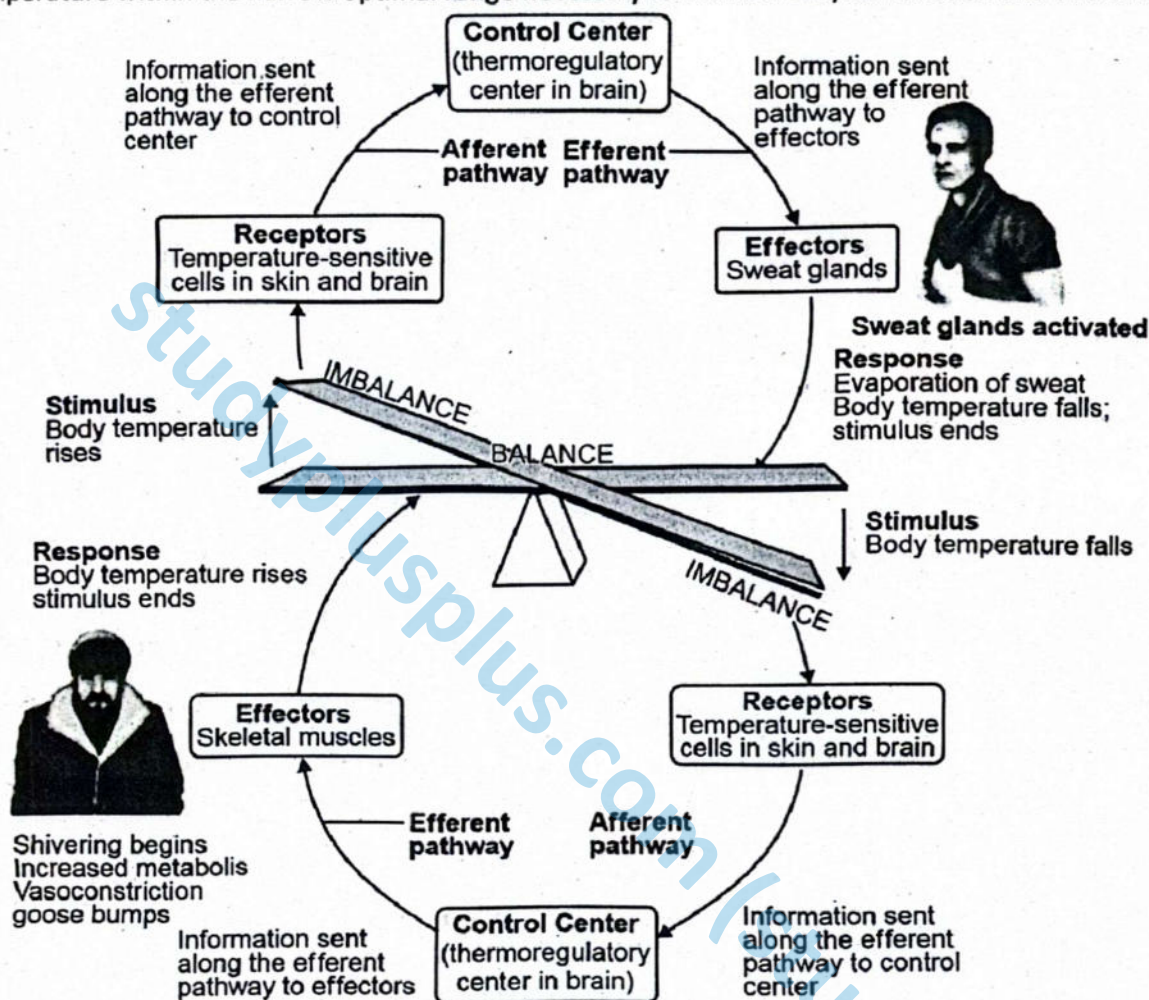


Figure: Thermoregulation in human

SLOs Based (MCQs)

THERMOREGULATION

- | | |
|---|---|
| <p>1. Which part of the brain functions as the body's thermostat?</p> <p>A. Cerebellum B. Medulla oblongata</p> <p>C. Hypothalamus D. Pituitary gland</p> <p>2. Thermoreceptors in the skin detect:</p> <p>A. Core temperature changes</p> <p>B. External temperature changes</p> <p>C. Blood pressure</p> <p>D. Blood glucose levels</p> <p>3. Which response involves widening of arterioles near the skin surface?</p> | <p>A. Vasoconstriction B. Piloerection</p> <p>C. Sweating D. Vasodilation</p> <p>4. What is the main method of heat loss through sweating?</p> <p>A. Radiation B. Conduction</p> <p>C. Evaporation D. Convection</p> <p>5. In dry environments, humans can tolerate temperatures up to:</p> <p>A. 37 °C B. 45 °C</p> <p>C. 56 °C D. 65 °C</p> |
|---|---|

6. Shivering helps to:
A. Reduce body temperature
B. Produce heat through muscle contractions
C. Cool the body via evaporation
D. Increase oxygen levels
7. Vasoconstriction during cold exposure helps to:
A. Increase sweating
B. Increase blood flow to the skin
C. Decrease heat loss
D. Lower body temperature
8. Goosebumps are caused by:
A. Muscle fatigue
B. Erector pili muscle contraction
C. Increased sweat production
D. Hormonal stimulation
9. Which hormone increases metabolic rate during cold exposure?
A. Insulin
B. Glucagon
C. Adrenaline
D. Estrogen
10. Shunt vessels help conserve heat by:
A. Increasing blood pressure
B. Directing blood away from the skin surface
C. Promoting sweating
D. Stimulating vasodilation
11. Which physiological response reduces heat loss in cold environments?
A. Sweating
B. Vasodilation
C. Shivering
D. Vasoconstriction
12. What is the effect of high humidity on sweat evaporation?
A. Increases efficiency
B. No effect
C. Decreases efficiency
D. Accelerates metabolism
13. The hypothalamus responds to changes in:
A. Ambient light
B. Skin pH
C. Blood temperature
D. Oxygen saturation
14. Which of the following is not a function of the hypothalamus in thermoregulation?
A. Detecting body temperature
B. Initiating shivering
C. Secreting insulin
D. Controlling vasodilation
15. Piloerection in humans is considered:
A. An active heat production mechanism
B. A behavioral response
C. A vestigial response
D. A hormonal process
16. Thermoregulation is essential to:
A. Maintain blood pressure
B. Ensure proper enzyme activity
C. Regulate heart rate
D. Control reflexes
17. Which of the following processes contributes directly to heat gain?
A. Sweating
B. Vasodilation
C. Shivering
D. Goosebumps
18. In cold conditions, blood is diverted away from the skin via:
A. Arteries
B. Veins
C. Shunt vessels
D. Lymph vessels
19. Sweat is secreted primarily by:
A. Sebaceous glands
B. Adrenal glands
C. Endocrine glands
D. Sweat glands
20. Which is a behavioral response to thermal discomfort?
A. Vasoconstriction
B. Piloerection
C. Wearing warm clothing
D. Shivering

Answer Key

1. C	2. B	3. D	4. C	5. D	6. B	7. C	8. B	9. C	10. B
11. D	12. C	13. C	14. C	15. C	16. B	17. C	18. C	19. D	20. C

Short Questions

Q.1 What role does the hypothalamus play in thermoregulation?

Ans. The hypothalamus is the central regulator of body temperature. It continuously monitors core temperature by detecting the warmth or coldness of blood flowing through it. It also integrates input from thermoreceptors in the skin that sense external conditions. Based on this information, the hypothalamus initiates physiological responses, such as sweating, vasodilation, or shivering, and behavioral responses like seeking shade or adding clothing, to maintain thermal balance.

MARKING SCHEME (3 Marks)

- Monitoring blood temperature – 1 mark; receiving input from receptors – 1 mark; initiating responses – 1 mark



Q.2 How does vasodilation help cool the body?

Ans. Vasodilation is the widening of blood vessels, particularly arterioles near the skin surface. This process increases blood flow to the skin, allowing more heat to be transferred from the body's core to the environment. Excess heat is then lost mainly through radiation and convection, effectively lowering body temperature.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; effect of increased blood flow – 1 mark; heat loss by radiation/convection – 1 mark

Q.3 Why is sweating an effective cooling mechanism?

Ans. Sweating regulates temperature through evaporative cooling. Sweat glands secrete water onto the skin surface, and as this liquid evaporates, it absorbs latent heat energy from the body. This heat transfer reduces core temperature, making sweating particularly effective in dry environments where evaporation occurs rapidly.

MARKING SCHEME (3 Marks)

- Sweating mechanism – 2 marks; cooling effect – 1 mark

Q.4 How does humidity affect the efficiency of sweating?

Ans. Humidity directly influences the evaporation of sweat. In high humidity, the air is already saturated with water vapor, slowing sweat evaporation and reducing its cooling effect. In contrast, dry air allows rapid evaporation, enhancing heat loss and making sweating a more effective thermoregulatory mechanism.

MARKING SCHEME (3 Marks)

- Relationship of humidity and evaporation – 2 marks; resulting effect on cooling – 1 mark

Q.5 What is shivering, and how does it aid in thermoregulation?

Ans. Shivering is the rapid, involuntary contraction and relaxation of skeletal muscles. These muscle movements increase metabolic activity, which generates extra heat as a byproduct of cellular respiration. This additional heat production helps restore core body temperature in cold environments, preventing hypothermia.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; role in generating heat – 2 marks

Q.6 Describe the function of vasoconstriction in cold environments.

Ans. Vasoconstriction is the narrowing of arterioles that supply blood to the skin. By reducing blood flow to surface capillaries, this mechanism limits heat loss through radiation, convection, and conduction at the skin surface. As a result, more heat is retained in the body's core, helping to maintain an optimal internal temperature during cold exposure.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; role in limiting heat loss – 1 mark; retaining core heat – 1 mark

Q.7 What is the significance of piloerection in thermoregulation?

Ans. Piloerection is the reflex action where small muscles at the base of hair follicles contract, causing hairs to stand upright. In fur-covered animals, this traps a layer of insulating air close to the skin, reducing heat loss. In humans, however, it serves little thermoregulatory purpose and is considered a vestigial response, though it signals exposure to cold.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; thermoregulatory role in animals – 1 mark; limited role in humans – 1 mark

Q.8 Describe how shunt vessels contribute to heat conservation.

Ans. Shunt vessels are specialized blood pathways that divert blood away from surface capillaries toward deeper veins. This redirection reduces heat loss from the skin by minimizing radiation and convection at the surface. Consequently, more heat is conserved in the body's core, making this mechanism especially useful in cold weather to prevent hypothermia.

MARKING SCHEME (3 Marks)

- Role of shunt vessels – 2 marks; effect of conserving core heat – 1 mark

Q.9 How does the body respond hormonally to cold exposure?

Ans. In cold environments, the hypothalamus stimulates the release of hormones such as adrenaline and noradrenaline. These hormones increase metabolic rate by accelerating cellular respiration, which generates additional heat. Thyroxine may also be released to boost long-term heat production. Collectively, these hormonal adjustments help maintain body temperature under prolonged cold stress.

MARKING SCHEME (3 Marks)

- Naming hormones (e.g., adrenaline / thyroxine) – 1 mark; functions in raising metabolism and heat production – 2 marks



Q.10 Why is thermoregulation crucial for enzyme function?

Ans. Enzymes function effectively only within a narrow temperature range. If body temperature falls too low, enzyme activity slows, reducing metabolic reactions. Conversely, excessive heat can denature enzymes by altering their active site, stopping reactions altogether. Thermoregulation maintains body temperature within the optimal range, ensuring stable enzyme activity and efficient metabolism.

MARKING SCHEME (3 Marks)

- Effects of temperature change on enzymes – 2 marks; role of thermoregulation in stability – 1 mark

SLO [B-12-R-78]

Describe three elements i.e., receptors, control center and effectors which operate homeostatic mechanisms.

HOMEOSTASIS AND ITS ROLE IN THERMOREGULATION

Homeostasis is the process by which living organisms maintain a stable internal environment despite constant changes in the external surroundings. It ensures that essential conditions such as temperature, pH, water balance, and glucose concentration remain within narrow limits necessary for survival. In humans, homeostasis is critical because enzymes and cellular processes can function properly only under stable internal conditions.

One of the most important examples of homeostasis is **thermoregulation**—the regulation of body temperature. For humans, the normal internal body temperature is around **36.5–37.5 °C**. If this range is disturbed, enzymes may lose efficiency, cells may suffer damage, and vital organs such as the brain and heart may fail to function correctly. To prevent such dangers, the body uses a coordinated system of **receptors, a control center, and effectors** to detect changes, process information, and restore balance.

Elements of Homeostatic Mechanisms

Homeostasis in humans and other animals is achieved through a highly coordinated system that involves three essential components: **receptors (sensors), a control center (integrator), and effectors (responders)**. These three elements form a feedback loop that continuously monitors the internal environment, detects changes, and restores balance whenever necessary. Together, they ensure that vital conditions such as temperature, pH, water balance, and glucose concentration remain within safe limits.

1. Receptors (Sensors)

Receptors are specialized cells or structures that detect changes, known as **stimuli**, in the body's internal or external environment. Their primary role is to monitor conditions and immediately send information to the control center.

In thermoregulation, two major types of receptors are involved:

- **Peripheral Thermoreceptors:** Located in the skin, these receptors sense changes in the external environment. For example, when the skin comes in contact with cold air or hot surfaces, the peripheral thermoreceptors detect these variations and send signals to the brain. This allows the body to prepare early responses even before the core temperature is significantly affected.
- **Central Thermoreceptors:** Found mainly in the hypothalamus, these receptors monitor the temperature of the blood as it circulates through the brain. Since blood temperature reflects the body's core temperature, central receptors provide the most accurate information about internal conditions.

Both sets of receptors work together: peripheral receptors act as an early-warning system for external changes, while central receptors provide precise monitoring of internal stability. This continuous flow of information ensures rapid detection of any deviation from the set point, allowing corrective measures to be initiated without delay.

2. Control Center (Integrator)

The **control center** is the decision-making component of the homeostatic system. It receives information from receptors, compares it with the body's normal set point, and initiates responses to correct any imbalance.



In humans, the **hypothalamus**—a small but crucial part of the brain—serves as the main control center for thermoregulation. The hypothalamus contains specialized nerve cells that function like a thermostat, constantly comparing incoming temperature data with the ideal set point of about **37 °C**.

- If the body temperature rises above normal, the hypothalamus detects this change and activates cooling mechanisms. These include stimulation of sweat glands for evaporative cooling, dilation of skin blood vessels (vasodilation) to increase heat loss, and a reduction in metabolic rate to limit internal heat production.
- If the body temperature falls below normal, the hypothalamus triggers heat-gain mechanisms. These include shivering of muscles to generate heat, vasoconstriction to conserve core temperature, and stimulation of endocrine glands (thyroid and adrenal) to increase metabolic heat production.

Thus, the hypothalamus not only processes sensory information but also coordinates appropriate responses by signaling the effectors. Without the integrative function of the control center, homeostasis would break down, and the body would be unable to adjust to environmental fluctuations.

3. Effectors (Responders)

Effectors are the structures that carry out the corrective actions determined by the control center. They receive instructions via nerve impulses or hormones and bring about changes that restore the internal environment to normal.

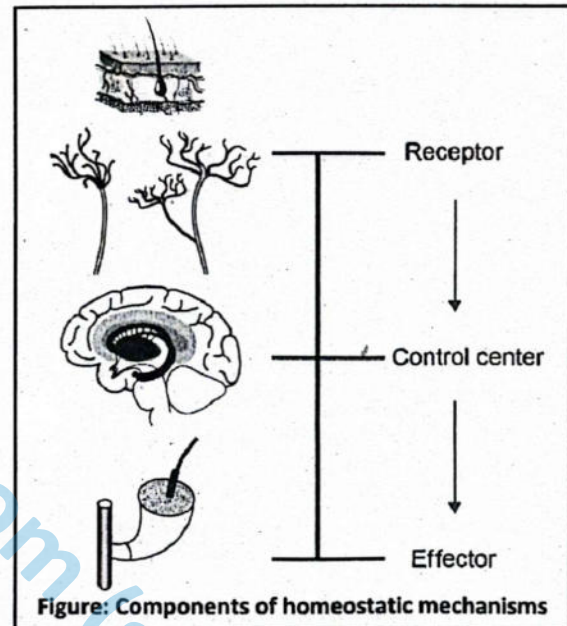
In thermoregulation, the key effectors include:

- **Sweat Glands:** Activated by the hypothalamus in hot conditions, sweat glands secrete fluid onto the skin surface. As the sweat evaporates, it removes latent heat from the body, producing a cooling effect.
- **Skin Arterioles:** These small blood vessels adjust their diameter in response to temperature changes. They **dilate** (vasodilation) to increase blood flow and heat loss in hot weather, and **constrict** (vasoconstriction) to reduce heat loss in cold conditions.
- **Skeletal Muscles:** When stimulated by the hypothalamus in cold conditions, skeletal muscles contract rapidly in shivering. These contractions generate heat through increased cellular respiration, warming the body.
- **Endocrine Glands:** The thyroid gland releases thyroxine, which raises the metabolic rate and heat production, while the adrenal medulla secretes adrenaline to increase energy release and body warmth during cold stress.

By executing these physiological changes, effectors directly restore the body's temperature to its normal range. Their role is crucial because they translate the brain's instructions into visible and measurable responses that protect the body from damage caused by extreme heat or cold.

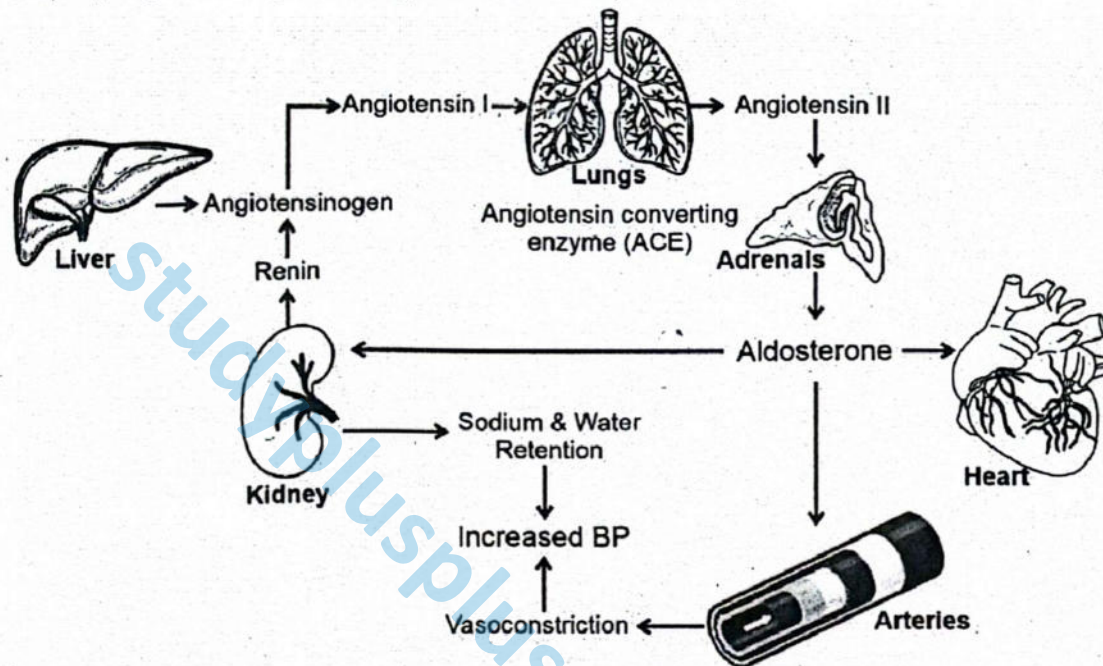
Homeostasis is the maintenance of a stable internal environment, vital for the survival of living organisms.

Thermoregulation is a prime example of this process, keeping body temperature within safe limits. The three elements of homeostatic mechanisms are: **receptors**, which detect changes in the environment; the **control center**, which processes the information and decides on corrective action; and **effectors**, which carry out the responses to restore balance. Together, these elements form a feedback system that preserves internal stability and supports life.



FOR YOUR INFORMATION

The renin-angiotensin system (RAS), also known as the renin-angiotensin aldosterone system (RAAS), is a critical hormonal system that regulates blood pressure and fluid balance. It is the system of hormones, proteins, enzymes and reactions. It regulates your blood pressure by increasing sodium (salt) reabsorption, water reabsorption (retention) and vascular tone (the degree to which your blood vessels constrict, or narrow). The RAAS consists of three major substances, including Renin (an enzyme), Angiotensin II (a hormone), and Aldosterone (a hormone).



SLO [B-12-R-79]

Relate the homeostatic mechanisms with the negative and positive feedback systems.

HOMEOSTATIC MECHANISMS AND FEEDBACK SYSTEMS

Homeostasis operates through feedback systems, which are self-regulating processes that maintain internal stability in response to external or internal changes. Feedback can be of two types: **negative feedback** and **positive feedback**. Both play vital roles in regulating physiological processes, but they differ in how they influence the original stimulus.

1. Negative Feedback Systems

Negative feedback is the most common and vital type of feedback in homeostatic regulation. In this system, the response generated by effectors **opposes or reverses** the initial change, thereby restoring balance to the normal set point.

► Role in Homeostasis

Most homeostatic mechanisms in the human body, including **thermoregulation**, **blood sugar control**, and **water balance**, are governed by negative feedback. It prevents extreme deviations and maintains conditions within **narrow limits** essential for enzyme activity and cellular function.



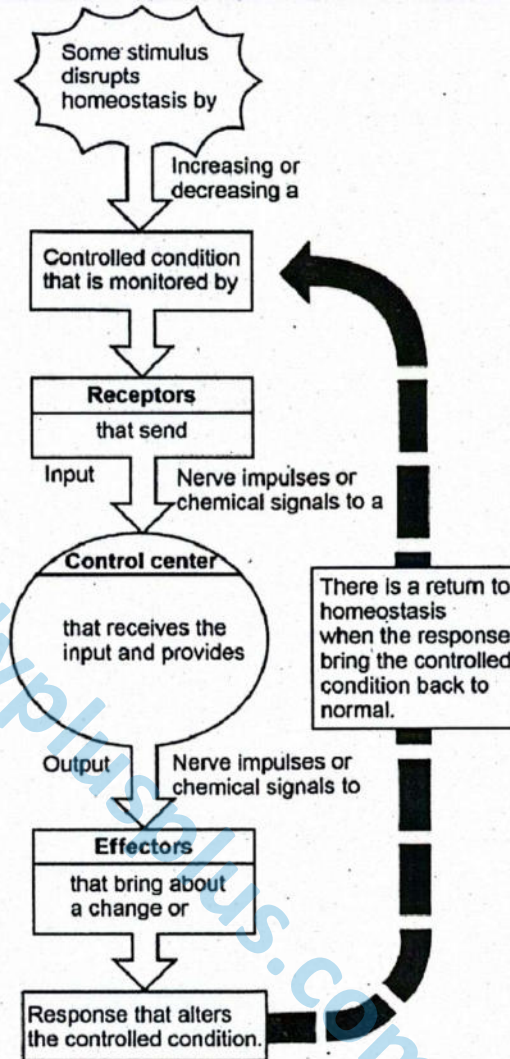


Figure: Negative Feedback Flowsheet

Example in Thermoregulation:

- **Rise in Temperature:** If body temperature increases above 37°C , thermoreceptors send signals to the hypothalamus. The hypothalamus activates sweating and vasodilation to reduce heat. As the body cools, the initial rise in temperature is corrected.
- **Fall in Temperature:** If body temperature drops below normal, thermoreceptors alert the hypothalamus. The hypothalamus triggers shivering, vasoconstriction, and hormonal changes to produce and conserve heat. As the body warms, the drop in temperature is reversed.
- **Regulation of Blood Glucose:** After food intake, blood glucose levels rise. This increase is detected by β -cells of the pancreas, which release insulin, promoting glucose uptake by cells and storage as glycogen in the liver. As glucose is removed from the blood, levels return to normal.
- Conversely, if blood glucose levels fall, α -cells of the pancreas release glucagon, which stimulates glycogen breakdown, raising blood glucose. Both responses restore balance through opposing actions, exemplifying negative feedback.
- **Regulation of Blood Calcium:** When blood calcium levels decline, the parathyroid glands secrete parathyroid hormone (PTH). PTH increases calcium absorption in the intestines and kidneys, and promotes calcium release from bones. Once calcium levels rise to the set point, PTH secretion is suppressed, completing the negative feedback loop.

In both cases, the response negates the original disturbance, exemplifying negative feedback.

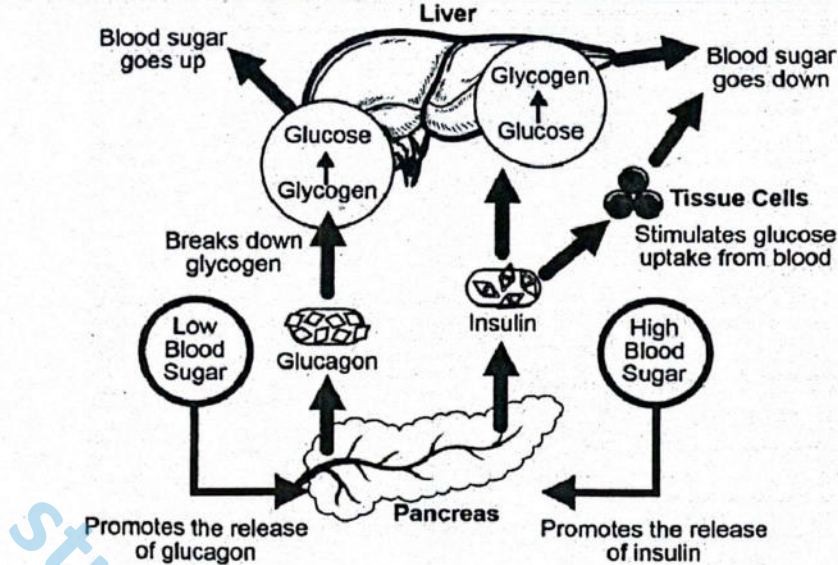


Figure: Blood sugar levels are controlled by a negative feedback flowsheet

2. Positive Feedback Systems

Positive feedback is less common in homeostasis. In this system, the response generated by effectors **amplifies or intensifies** the original stimulus, pushing the variable further away from the set point. This loop continues until an external factor interrupts it.

► Role in the Body

Positive feedback is usually associated with processes that need a rapid, decisive outcome rather than ongoing balance. While it is not the main mechanism of homeostasis, it can temporarily override normal stability for specific biological purposes.

Examples:

- **Blood Clotting:** When a blood vessel is injured, platelets adhere to the damaged site and release chemical signals that attract additional platelets. This process accelerates until a **sufficient clot is formed** to seal the wound. Once the injury is resolved, the feedback loop terminates.
- **Uterine Contractions During Childbirth:** The hormone **oxytocin**, released by the **posterior pituitary**, stimulates uterine contractions. The stretching of the cervix is sensed by the nervous system, which signals for increased oxytocin secretion. This amplifies contractions until childbirth occurs, at which point the stimulus (cervical stretch) ceases and the loop ends.

In both cases, the response amplifies the stimulus until the process reaches a natural endpoint.

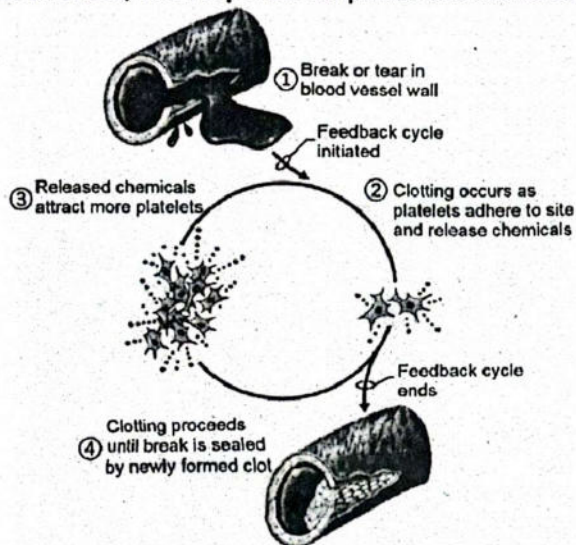


Figure: Blood Clotting Mechanism during positive feedback system.

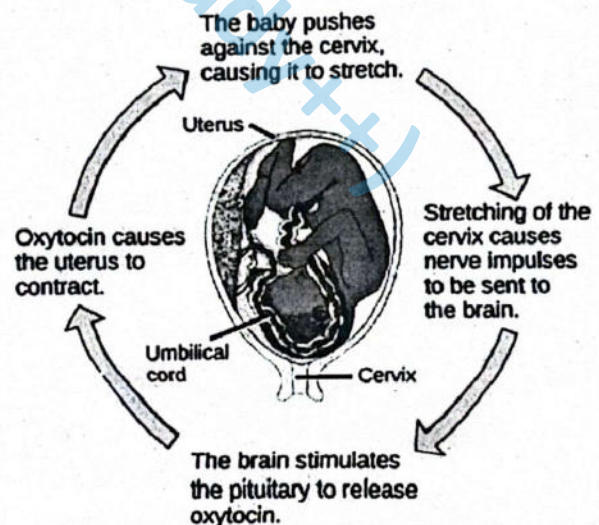


Figure: Positive feedback loop: The birth of a human infant is the result of positive feedback

3. Comparison and Relationship with Homeostasis

- **Negative feedback** maintains stability by counteracting deviations (e.g., regulating temperature, glucose, water, and blood pressure). It is essential for survival and long-term balance.
 - **Positive feedback** temporarily enhances changes to achieve a specific outcome (e.g., childbirth, blood clotting). While not maintaining stability in the short term, it ultimately serves the body's survival.
- Thus, homeostatic mechanisms primarily rely on **negative feedback** to sustain equilibrium, but **positive feedback** can complement them in special situations requiring a rapid and decisive result.

Comparison of Negative and Positive Feedback Mechanisms in Homeostasis

Feature	Negative Feedback Mechanism	Positive Feedback Mechanism
Definition	A regulatory mechanism in which the response reverses the direction of the original stimulus to restore balance.	A regulatory mechanism in which the response amplifies or reinforces the original stimulus.
Role in Homeostasis	Maintains homeostasis by keeping internal conditions within a narrow range.	Temporarily disrupts homeostasis to achieve a specific biological outcome.
Direction of Response	Opposes the initial change.	Enhances the initial change.
Stability Effect	Promotes internal stability and equilibrium.	Drives the system away from equilibrium until a specific endpoint is reached.
Commonness	Most common form of feedback in biological systems.	Less common, used in specialized situations.
Duration	Operates continuously to regulate internal variables.	Operates temporarily until the process is completed.
Examples	Regulation of blood glucose, body temperature, blood pressure, and calcium levels.	Blood clotting, uterine contractions during childbirth, lactation reflex.
Involvement of Set Point	Aims to maintain or restore a physiological set point.	Temporarily overrides the set point until the process ends.
Type of Feedback Loop	Corrective or stabilizing loop.	Amplifying or self-reinforcing loop.
System Outcome	Returns the system to its normal state.	Drives the system to a new state or conclusion.

SLOs Based (MCQs)

HOMEOSTASIS

1. **What is homeostasis?**
 - A. Ability to respond to stimuli
 - B. Maintenance of a stable internal environment
 - C. Increase in body temperature
 - D. Breakdown of complex substances
2. **The internal environment of the body primarily includes:**
 - A. Bone marrow and cartilage
 - B. Muscle fibers and ligaments
 - C. Interstitial fluid and plasma
 - D. Air in the lungs
3. **Which two systems coordinate to maintain homeostasis?**
 - A. Circulatory and muscular
 - B. Nervous and endocrine
 - C. Digestive and reproductive
 - D. Skeletal and integumentary
4. **What is the function of receptors in a homeostatic mechanism?**
 - A. To initiate movement
 - B. To receive and detect changes in environment
 - C. To control hormone levels
 - D. To store nutrients
5. **The control center in a feedback loop:**
 - A. Produces enzymes
 - B. Carries out a response
 - C. Processes information and sends commands
 - D. Absorbs nutrients
6. **The effector in a homeostatic system:**
 - A. Detects stimuli
 - B. Processes signals
 - C. Carries out the response
 - D. Monitors hormones



7. **Negative feedback results in:**
 A. Acceleration of a process
 B. Further deviation from normal
 C. Restoration of balance
 D. Disruption of equilibrium
8. **Which of the following is an example of negative feedback?**
 A. Uterine contractions during labor
 B. Blood clot formation
 C. Regulation of blood glucose
 D. Lactation reflex
9. **Which hormone lowers blood glucose levels after a meal?**
 A. Glucagon
 B. Thyroxine
 C. Adrenaline
 D. Insulin
10. **Glucagon is released when:**
 A. Blood glucose is high
 B. Oxygen levels increase
 C. Blood glucose is low
 D. Body temperature rises
11. **Positive feedback differs from negative feedback because it:**
 A. Inhibits biological processes
 B. Restores normal conditions
 C. Amplifies the original stimulus
 D. Maintains homeostasis directly
12. **Which of the following is a positive feedback mechanism?**
 A. Regulation of pH
 B. Blood clotting
 C. Thermoregulation
 D. Calcium balance
13. **Oxytocin stimulates:**
 A. Glucose absorption
 B. Blood pressure drop
 C. Uterine contractions
 D. Muscle fatigue
14. **Which structure is involved in glucose regulation?**
 A. Heart
 B. Pancreas
 C. Kidneys
 D. Lungs
15. **The renin-angiotensin system regulates:**
 A. Blood sugar
 B. Blood calcium
 C. Blood pressure
 D. Blood clotting
16. **Blood plasma is part of the:**
 A. External environment
 B. Digestive tract
 C. Internal environment
 D. Skeletal system
17. **Which organ converts glucose to glycogen after a meal?**
 A. Kidney
 B. Liver
 C. Pancreas
 D. Small intestine
18. **Parathyroid hormone raises blood calcium by acting on the:**
 A. Liver
 B. Pancreas
 C. Bones, intestines, and kidneys
 D. Muscles and lungs
19. **Which of the following systems directly supports gas exchange in homeostasis?**
 A. Digestive system
 B. Nervous system
 C. Respiratory system
 D. Integumentary system
20. **Feedback mechanisms function as:**
 A. One-time responses
 B. Linear processes
 C. Self-regulating loops
 D. Immune responses

Answer Key

1. B	2. C	3. B	4. B	5. C	6. C	7. C	8. C	9. D	10. C
11. C	12. B	13. C	14. B	15. C	16. C	17. B	18. C	19. C	20. C

Short Questions

Q.1 Define homeostasis and describe its significance.

Ans. Homeostasis is the ability of an organism to maintain a stable internal environment despite changes in the external surroundings. It regulates variables such as temperature, pH, and glucose concentration within narrow limits. This stability is vital because enzymes function optimally only under specific conditions, and disruption of balance can impair cellular activities. By maintaining homeostasis, organisms ensure proper metabolism, survival, and adaptation to environmental changes.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; significance – 1 mark

Q.2 What are the main components of a homeostatic feedback mechanism?

Ans. A homeostatic feedback mechanism is made up of three key components. The receptor detects any change in the internal or external environment. The control center, usually the brain or endocrine glands, processes this information and decides the corrective response. The effector then carries out the response, restoring the system to its equilibrium or set point. Together, these components ensure stability within the organism.

MARKING SCHEME (3 Marks)

- Receptor – 1 mark; control center – 1 mark; effector – 1 mark



Q.3 Differentiate between negative and positive feedback mechanisms.

Ans. Negative feedback mechanisms reverse deviations from a set point, restoring the system to balance. For example, regulation of blood glucose and body temperature both rely on negative feedback. Positive feedback mechanisms, on the other hand, amplify changes and drive processes toward completion rather than stabilization. Classic examples include blood clotting and uterine contractions during childbirth. Both mechanisms are crucial for physiological regulation, but they operate differently depending on the body's needs.

MARKING SCHEME (3 Marks)

- Negative feedback definition – 1.5 marks; positive feedback definition – 1.5 marks

Q.4 Explain how insulin and glucagon maintain blood glucose levels.

Ans. When blood glucose rises after eating, the pancreas releases insulin. Insulin lowers glucose levels by stimulating cellular uptake of glucose and promoting its storage as glycogen in the liver and muscles. When glucose levels fall between meals, the pancreas secretes glucagon. Glucagon raises glucose levels by stimulating glycogen breakdown and glucose release from the liver. This coordinated action of insulin and glucagon is a negative feedback system that maintains glucose homeostasis.

MARKING SCHEME (3 Marks)

- Role of insulin – 1.5 marks; role of glucagon – 1.5 marks

Q.5 How does positive feedback work during childbirth?

Ans. During labor, stretching of the uterus stimulates the release of the hormone oxytocin from the pituitary gland. Oxytocin increases the intensity and frequency of uterine contractions, which further stretches the uterus. This additional stretching triggers more oxytocin release, creating a cycle of increasing contractions. The loop continues until the baby is delivered. Once birth occurs, the stimulus ends, and the positive feedback cycle is terminated.

MARKING SCHEME (3 Marks)

- Name of hormone – 1 mark; role in amplifying contractions – 2 marks

SLO [B-12-R-81]

Define osmoregulation.

OSMOREGULATION

Osmoregulation is the biological process through which living organisms regulate the balance of water and dissolved solutes—such as salts, ions, and other osmolytes—in their body fluids. It ensures that the osmotic pressure inside cells and tissues remains stable, preventing harmful effects of excessive water gain (swelling) or water loss (shrinkage).

Importance

Maintaining osmotic balance is essential because all metabolic activities occur in an aqueous medium. If water and solute concentrations fluctuate widely, enzyme function, cellular transport, and overall metabolism are disrupted. Osmoregulation therefore plays a vital role in maintaining **homeostasis**, ensuring that cells function efficiently regardless of external environmental changes.

How It Works?

Osmoregulation involves three main components:

1. **Receptors (Osmoreceptors):** Specialized sensors, mainly in the hypothalamus, detect changes in water concentration and osmotic pressure of blood plasma.
2. **Control Center:** The hypothalamus processes this information and, if needed, signals the pituitary gland to release antidiuretic hormone (ADH).
3. **Effectors:** The kidneys act as the primary effectors. They adjust the volume and concentration of urine by either conserving water when the body is dehydrated or excreting excess water when the body is overhydrated.

Examples:

- In humans, when water levels drop (e.g., due to sweating), ADH secretion increases. This makes kidney tubules reabsorb more water, producing concentrated urine.
- When water intake is high, ADH secretion decreases, leading to dilute urine and restoring osmotic balance.



- In aquatic animals, such as freshwater fishes, osmoregulation prevents excess water entry, while in marine fishes, it prevents dehydration caused by salty surroundings.

In simple terms, osmoregulation is the **control of water and solute balance** in the body. It is crucial for maintaining stable internal conditions, supporting cellular activities, and ensuring survival in environments with fluctuating water and salt concentrations.

SLO [B-12-R-80]

Differentiate between osmoconformers and osmoregulators.

OSMOCONFORMERS AND OSMOREGULATORS

Osmoconformers

Osmoconformers are organisms whose internal osmotic concentration changes in accordance with that of their surrounding environment. Their body fluids are **isotonic** to the external medium, meaning there is no significant osmotic gradient between the internal and external fluids. As a result, these organisms do not expend significant energy on active osmoregulation.

- Most **marine invertebrates** (e.g., mollusks, echinoderms), some **freshwater invertebrates**, and certain **marine vertebrates**, such as **hagfishes (Myxine)** and **elasmobranchs** (sharks and rays), are classified as osmoconformers.
- In elasmobranchs, isotonicity is achieved through the accumulation of organic solutes in their blood.

Osmoregulators

Osmoregulators are organisms that actively maintain a constant internal osmotic concentration of body fluids, irrespective of changes in the osmolarity of their external environment. This regulation is essential for sustaining cellular function and overall homeostasis.

Organisms living in **freshwater, marine, and terrestrial environments** face different osmotic challenges. In response, osmoregulators have evolved a variety of **physiological and structural adaptations** that allow them to regulate the movement of water and solutes across body surfaces and membranes.

Such adaptations may include:

- **Excretory structures** (e.g., kidneys, nephridia, Malpighian tubules).
- **Water-proof body coverings** (e.g., scales, exoskeletons, keratinized skin).
- **Active transport systems** in gills or epithelial tissues for salt uptake or excretion.
- **Behavioral mechanisms** like burrowing or nocturnal activity to reduce water loss.

These adaptations vary based on the **habitat type**:

- **Freshwater animals** prevent excessive water uptake and conserve salts.
- **Marine animals** combat dehydration and excessive salt intake.
- **Terrestrial animals** minimize water loss while maintaining salt balance.

SCIENCE TITBITS

In most vertebrates, the level of urea to this high concentration would damage the proteins because it is chaotropic (denaturing) agent that disrupts noncovalent and ionic bonds between amino acids residues, but the presence of TMAO helps to stabilize these protein molecules against the adverse effects of urea.

Feature	Osmoregulation	Osmoconformation
Definition	The active regulation of internal osmotic concentration, independent of the environment.	The passive adjustment of internal osmotic concentration to match the environment.
Mechanism	Involves energy-dependent physiological processes to maintain homeostasis.	Involves minimal energy expenditure; osmolarity changes with the environment.
Internal Environment	Maintained constant regardless of external osmotic conditions.	Fluctuates with changes in the external osmotic environment.
Adaptation	Allows survival in diverse and fluctuating environments (freshwater, marine, <i>terrestrial</i>).	Limited to stable osmotic environments like marine habitats.
Examples	Freshwater fish, marine vertebrates (except elasmobranchs), amphibians, humans.	Marine invertebrates, elasmobranchs (sharks, rays), hagfish.



Osmotic Relationship with Environment	Often hypertonic or hypotonic relative to the environment.	Usually isotonic with their environment.
Use of Osmolytes	May actively excrete or retain ions and water to regulate internal conditions.	May accumulate organic osmolytes like urea and TMAO to balance osmotic pressure.
Energy Requirement	High, due to active transport of solutes and water.	Low, since they rely on passive equilibrium.

SLO [B-12-R-82]

Explain the problems faced by osmoregulators.

PROBLEMS FACED BY OSMOREGULATORS

Osmoregulators actively maintain a constant internal balance of water and solutes, regardless of external environmental conditions. While this ability allows them to survive in diverse habitats, it also presents several challenges. These problems differ depending on whether the animal lives in **freshwater, marine, or terrestrial environments**.

1. Freshwater Osmoregulators

Freshwater animals live in an environment where the concentration of salts outside is much lower than inside their body fluids. This creates a constant osmotic challenge.

Problems Faced:

- **Water Influx by Osmosis:** Water continuously enters the body across permeable surfaces (skin or gills). This can cause cells to swell and burst if not regulated.
- **Loss of Salts:** Essential ions like sodium and chloride tend to diffuse out of the body into the dilute environment.

Consequences: Without regulation, freshwater animals would become diluted and lose critical electrolytes.

2. Marine Osmoregulators

Marine animals live in seawater, which has a much higher salt concentration than their body fluids. This creates the opposite osmotic stress compared to freshwater animals.

Problems Faced:

- **Loss of Water by Osmosis:** Water tends to leave the body, leading to dehydration.
- **Excess Salt Gain:** Salt enters the body across gill membranes and from drinking seawater.

Consequences: Without regulation, marine osmoregulators would become dehydrated and accumulate toxic levels of salts.

3. Terrestrial Osmoregulators

Land animals face challenges due to exposure to air and a relatively dry environment.

Problems Faced:

- **Water Loss by Evaporation:** Evaporation from skin, respiratory surfaces, and excretion can rapidly deplete body water.
- **Limited Water Availability:** Many terrestrial habitats have scarce or irregular sources of fresh water.
- **Salt Balance:** Sweating and excretion can lead to loss of essential electrolytes.

Consequences: Without mechanisms to conserve water, terrestrial animals risk dehydration and impaired physiological functions.

4. Energy Demands of Osmoregulation

A common problem across all environments is the **high energy cost** of osmoregulation. Active transport of ions across membranes, production of concentrated or dilute urine, and specialized glandular secretions (e.g., salt glands in marine birds) all consume large amounts of ATP. This means osmoregulators must balance energy expenditure for survival with other biological needs like growth and reproduction.

Osmoregulators face significant problems depending on their habitat:

- **Freshwater animals** struggle with excess water gain and salt loss.
- **Marine animals** struggle with dehydration and salt overload.
- **Terrestrial animals** face water loss through evaporation and limited water supply.



Additionally, osmoregulation requires **continuous energy investment**, which can strain the animal's resources. Despite these challenges, specialized adaptations allow osmoregulators to survive in diverse environments.

SLO [B-12-R-83]

Explain the different methods of osmoregulation found in freshwater, marine water and terrestrial.

Methods of Osmoregulation in Different Habitats

Animals live in aquatic and terrestrial environments with very different osmotic pressures. To survive, they must regulate the balance of water and salts within their bodies. The strategies employed depend on whether the animal lives in **freshwater, marine, or terrestrial habitats**.

1. Osmoregulation in Freshwater Animals

Freshwater animals live in a **hypotonic environment**, where the concentration of solutes in the surrounding water is lower than that of their body fluids. Consequently, water continuously enters their bodies by osmosis, and they face a risk of salt loss.

Freshwater is a **hypotonic environment**, meaning the concentration of solutes outside the body is lower than inside body fluids. As a result, water continuously enters the body by osmosis, while essential salts tend to diffuse out. Without regulation, cells would swell and burst, and electrolytes would be depleted.

Adaptations of Freshwater Animals:

- **Excretion of Dilute Urine:** Freshwater fishes and amphibians produce large volumes of dilute urine, which removes excess water from the body.
- **Salt Reabsorption by Kidneys:** The kidneys actively reabsorb useful ions such as sodium and chloride to minimize their loss.
- **Active Salt Uptake:** Specialized cells called **ionocytes (chloride cells)**, located in the gills of fishes and the skin of amphibians, actively transport salts from the water into the body.
- **Dietary Compensation:** Salts lost to the environment are replaced through food intake.

Example: Most freshwater fishes (carps, tilapia) and amphibians (frogs) are **osmoregulators**, maintaining body fluids at higher solute concentrations than their environment.

2. Osmoregulation in Marine Animals

Marine teleosts (bony fishes) live in a **hypertonic environment**, where the concentration of solutes in seawater exceeds that of their internal body fluids. As a result, they tend to **lose water to the environment** by osmosis, particularly across the gill epithelium, and face an **accumulation of excess salts** due to ingestion of seawater.

Adaptations of Marine Teleosts (Bony Fishes):

- **Drinking Seawater:** Marine fishes drink seawater to replace water lost by osmosis.
- **Active Salt Excretion:**
 - **Monovalent ions** (Na^+ , Cl^- , K^+) are excreted through chloride cells in the gills.
 - **Divalent ions** (Mg^{2+} , Ca^{2+} , SO_4^{2-}) are excreted via the kidneys.
- **Specialized Salt Glands:** Some species possess **rectal glands** (e.g., in sharks) that eliminate excess salts into the digestive tract.
- **Partial Osmoconformation in Elasmobranchs:** Sharks and rays retain **urea and trimethylamine oxide (TMAO)** in their blood, making body fluids nearly isotonic with seawater and reducing osmotic stress.

Examples:

- Marine bony fishes like tuna and cod rely heavily on gill and kidney functions.
- Sharks use urea retention along with rectal glands.
- Marine birds (seagulls) and reptiles (sea turtles) use salt glands to expel concentrated saline solutions.

3. Osmoregulation in Terrestrial Animals

Terrestrial animals live in environments where **water is often scarce**, and they face constant risks of dehydration due to evaporation from skin, lungs, and excretion. Most are hypotonic relative to their surroundings, so preventing water loss is a major challenge.



Adaptations of Terrestrial Animals:

- **Structural Barriers:**
 - Arthropods possess a **chitinous exoskeleton** that reduces water loss.
 - Reptiles, birds, and mammals have **dead, keratinized skin layers** that act as waterproof barriers.
- **Kidney Function:** Mammalian kidneys produce concentrated urine. Desert mammals like the **kangaroo rat** have exceptionally long loops of Henle that conserve water efficiently.
- **Behavioral Adaptations:**
 - Many desert animals are **nocturnal**, avoiding the heat of the day.
 - Burrowing behavior reduces exposure to high temperatures.
- **Metabolic Water Production:** Some species obtain water from the breakdown of carbohydrates during respiration. For example, kangaroo rats meet up to **90% of their water requirements** from metabolic water.
- **Tolerance of Dehydration (Anhydrobiosis):** Certain animals can withstand extended periods without drinking water by storing fat or tolerating fluctuations in body water content.

Examples:

- Humans rely mainly on kidneys and hormonal regulation (ADH, aldosterone).
- Kangaroo rats survive almost entirely without drinking water.
- Camels tolerate dehydration and conserve water by fluctuating their body temperature, reducing sweating.

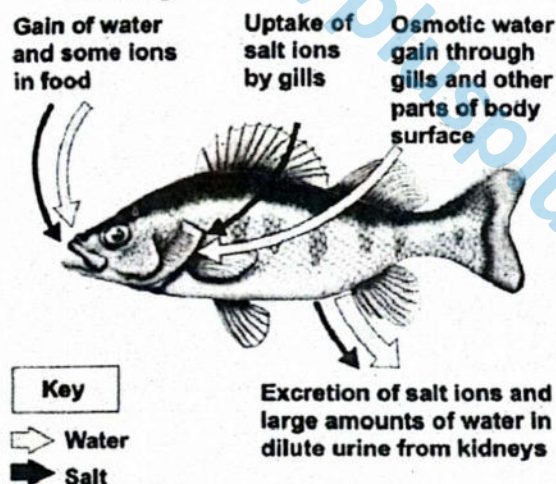


Figure: Osmoregulation in freshwater animals

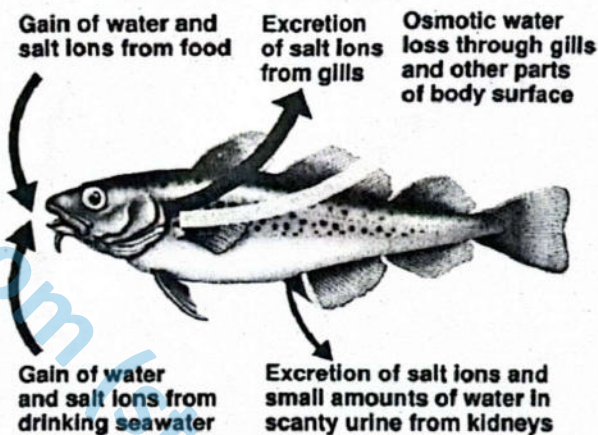


Figure: Osmoregulation in marine animals

Osmoregulatory Challenges and Adaptations in Different Habitats

Habitat	Osmoregulatory Problems	Adaptations
Freshwater	• Continuous water influx due to hypotonic environment • Loss of body salts by diffusion	• Excrete large volumes of dilute urine • Actively absorb ions (Na^+, Cl^-) through gills or skin • Possess impermeable body coverings
Marine	• Constant water loss due to hypertonic environment • Excessive salt gain from surroundings	• Drink seawater to compensate water loss • Excrete excess salts through salt glands or gill epithelium • Concentrated urine to conserve water
Terrestrial	• Desiccation due to evaporation • Limited water availability	• Impermeable integument (keratinized skin, waxy cuticle) • Efficient kidneys producing concentrated urine • Behavioral adaptations (e.g., burrowing, nocturnal activity)

OSMOREGULATION

1. **Osmoregulation refers to the regulation of:**
 - A. Enzyme activity
 - B. Oxygen concentration
 - C. Body temperature
 - D. Water and solute balance
2. **Animals that adjust their internal osmotic concentration to match the external environment are called:**
 - A. Osmoregulators
 - B. Osmoconformers
 - C. Homeotherms
 - D. Thermoregulators
3. **Which of the following animals is an example of an osmoconformer?**
 - A. Frogs
 - B. Sharks
 - C. Kangaroo rats
 - D. Freshwater bony fishes
4. **Osmoregulators differ from osmoconformers because they:**
 - A. Always remain hypertonic
 - B. Adjust internal osmolarity actively
 - C. Do not require energy
 - D. Are all marine vertebrates
5. **Ionocytes are specialized cells involved in:**
 - A. Digestive enzyme secretion
 - B. Thermoregulation
 - C. Active salt uptake
 - D. Hormone synthesis
6. **Freshwater animals typically face which of the following challenges?**
 - A. Water loss and salt gain
 - B. Water gain and salt loss
 - C. Water loss and salt loss
 - D. Water gain and salt gain
7. **The urine of freshwater animals is usually:**
 - A. Concentrated
 - B. Isotonic
 - C. Highly diluted
 - D. Absent
8. **In marine teleosts, excess salts are removed primarily through the:**
 - A. Skin
 - B. Lungs
 - C. Gill epithelium
 - D. Large intestine
9. **Rectal glands involved in salt secretion are found in:**
 - A. Amphibians
 - B. Marine reptiles
 - C. Some marine fishes
 - D. Freshwater invertebrates
10. **Which ions are typically excreted by the kidneys of marine fishes?**
 - A. Na^+ and Cl^-
 - B. Mg^{2+} and Ca^{2+}
 - C. H^+ and OH^-
 - D. Fe^{2+} and Cu^{2+}
11. **Marine teleosts drink seawater to:**
 - A. Eliminate salts
 - B. Maintain buoyancy
 - C. Compensate for water loss
 - D. Avoid predators
12. **What is the primary function of ionocytes in freshwater fishes?**
 - A. Secreting hormones
 - B. Preventing water absorption
 - C. Absorbing salts from the environment
 - D. Excreting nitrogenous waste
13. **Osmoconformers maintain:**
 - A. An osmotic concentration higher than their environment
 - B. A fixed internal solute composition
 - C. Internal osmotic balance equal to their environment
 - D. Water balance through metabolic water
14. **Which of the following is *not* an osmoregulatory adaptation of terrestrial animals?**
 - A. Keratinized skin
 - B. Chitinous exoskeleton
 - C. Dilute urine
 - D. Nocturnal behavior
15. **Anhydrobiosis refers to the ability to:**
 - A. Survive underwater
 - B. Store water in the body
 - C. Tolerate extreme dehydration
 - D. Produce very dilute urine
16. **Which organism is a classic example of anhydrobiosis?**
 - A. Camel
 - B. Tilapia
 - C. Shark
 - D. Kangaroo rat
17. **How do desert rodents obtain most of their water?**
 - A. By drinking water
 - B. By absorbing moisture from the air
 - C. From metabolic breakdown of food
 - D. Through skin absorption
18. **Which group of animals uses gill ionocytes to regulate salt balance?**
 - A. Reptiles
 - B. Amphibians and fishes
 - C. Mammals
 - D. Birds
19. **The primary osmotic challenge for marine teleosts is:**
 - A. Gaining too much water
 - B. Losing water and gaining excess salts
 - C. Gaining excess water and salts
 - D. Retaining metabolic waste



20. What feature helps arthropods reduce water loss in dry environments?
A. Internal salt glands

- B. Excretion of dilute urine
C. Chitinous exoskeleton
D. Secreting urea

Answer Key

1. D	2. B	3. B	4. B	5. C	6. B	7. C	8. C	9. C	10. B
11. C	12. C	13. C	14. C	15. C	16. D	17. C	18. B	19. B	20. C

Short Questions

Q.1 Define homeostasis and describe its significance.

Ans. Homeostasis is the ability of an organism to maintain a stable internal environment despite changes in the external surroundings. It regulates variables such as temperature, pH, and glucose concentration within narrow limits. This stability is vital because enzymes function optimally only under specific conditions, and disruption of balance can impair cellular activities. By maintaining homeostasis, organisms ensure proper metabolism, survival, and adaptation to environmental changes.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; significance – 1 mark

Q.2 What are the main components of a homeostatic feedback mechanism?

Ans. A homeostatic feedback mechanism is made up of three key components. The receptor detects any change in the internal or external environment. The control center, usually the brain or endocrine glands, processes this information and decides the corrective response. The effector then carries out the response, restoring the system to its equilibrium or set point. Together, these components ensure stability within the organism.

MARKING SCHEME (3 Marks)

- Receptor – 1 mark; control center – 1 mark; effector – 1 mark

Q.3 Differentiate between negative and positive feedback mechanisms.

Ans. Negative feedback mechanisms reverse deviations from a set point, restoring the system to balance. For example, regulation of blood glucose and body temperature both rely on negative feedback. Positive feedback mechanisms, on the other hand, amplify changes and drive processes toward completion rather than stabilization. Classic examples include blood clotting and uterine contractions during childbirth. Both mechanisms are crucial for physiological regulation, but they operate differently depending on the body's needs.

MARKING SCHEME (3 Marks)

- Negative feedback definition – 1.5 marks; positive feedback definition – 1.5 marks

Q.4 Explain how insulin and glucagon maintain blood glucose levels.

Ans. When blood glucose rises after eating, the pancreas releases insulin. Insulin lowers glucose levels by stimulating cellular uptake of glucose and promoting its storage as glycogen in the liver and muscles. When glucose levels fall between meals, the pancreas secretes glucagon. Glucagon raises glucose levels by stimulating glycogen breakdown and glucose release from the liver. This coordinated action of insulin and glucagon is a negative feedback system that maintains glucose homeostasis.

MARKING SCHEME (3 Marks)

- Role of insulin – 1.5 marks; role of glucagon – 1.5 marks

Q.5 How does positive feedback work during childbirth?

Ans. During labor, stretching of the uterus stimulates the release of the hormone oxytocin from the pituitary gland. Oxytocin increases the intensity and frequency of uterine contractions, which further stretches the uterus. This additional stretching triggers more oxytocin release, creating a cycle of increasing contractions. The loop continues until the baby is delivered. Once birth occurs, the stimulus ends, and the positive feedback cycle is terminated.

MARKING SCHEME (3 Marks)

- Name of hormone – 1 mark; role in amplifying contractions – 2 marks



I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. The main difference between endotherms and ectotherms is
 - A. how they conserve water
 - B. from where they get most of their body heat
 - C. whether they are cold or warm blooded
 - D. whether they live on land or water
2. Which word is not similarly with the others?
 - A. ectoderms
 - B. endoderms
 - C. homeothermic
 - D. warm-blooded
3. This part of the brain controls temperature regulation.
 - A. cerebellum
 - B. thalamus
 - C. hypothalamus
 - D. hippocampus
4. If body temperature decreases, which one of the following is incorrect?
 - A. vasoconstriction
 - B. hair raised
 - C. increase metabolic rate
 - D. sweating
5. The body's ability to maintain a constant body temperature is called what:
 - A. thermoregulation
 - B. modification
 - C. adaptation
 - D. lucky
6. The liver's functions are required for
 - A. thermoregulation
 - B. osmoregulation
 - C. homeostasis
 - D. all of these
7. The mechanism of regulation, generally between organism and its environment, of solutes and the gain and the loss of water is called
 - A. homeostasis
 - B. hemostasis
 - C. osmoregulation
 - D. thermoregulation
8. In a hot summer after noon, if your body's homeostatic machinery keep your internal temperature quite lower than that of external than this is an example of
 - A. positive feed back
 - B. negative feed back
 - C. feed back
 - D. osmoregulation
9. The characteristic which enable animals to tolerate dehydration is called
 - A. osmoconformation
 - B. osmoregulation
 - C. anhydrobiosis
 - D. anhidrosis
10. Homeostasis
 - A. must always be restored using negative feedback mechanisms
 - B. provides for the tight regulation of all physiological variables
 - C. is the sum of all chemical reactions in the body
 - D. is a combination of positive and negative feedback mechanisms
11. A negative feedback mechanism comprise
 - A. detectors, comparators, a fixed set point and effectors
 - B. detectors, amplifiers, comparators, a set point and effectors
 - C. detectors, attenuators, comparators, a set point and effectors
 - D. detectors, comparators, a variable set point and effectors
12. Positive feedback
 - A. does not exist in physiological systems
 - B. is seen in the initiation of an action potential when sodium entry causes depolarization which in turn leads to potassium exit
 - C. is unstable and requires some mechanism to break the feedback loop
 - D. is switched off by negative feedback mechanisms
13. Which of the following is an example of a positive feedback mechanism?
 - A. the control of blood sugar by insulin
 - B. blood clotting
 - C. the control of blood pressure
 - D. the regulation of body temperature
14. Which of the following components of a homeostatic regulatory mechanism is responsible for detecting a stimulus?
 - A. an effector
 - B. a control center
 - C. a receptor
 - D. an afferent pathway
15. Which of the following phrases most correctly defines homeostasis? Homeostasis involves the maintenance of internal conditions within _____.
 - A. specific set-points.
 - B. specific normal ranges.
 - C. ranges higher than external conditions.
 - D. ranges equal to external conditions.



16. Homeostasis is best conceived as a state of:
 A. negative feedback B. dynamic consistency
 C. positive feedback D. hormone regulation
17. Insulin is a hormone that acts to:
 A. stimulate metabolism in the body
 B. increase blood sugar levels
 C. decrease blood sugar levels

- D. stimulate the formation of blood clots
18. A setpoint is usually:
 A. the top of a normal range
 B. the bottom of a normal range
 C. in the middle of a normal range
 D. the point at which changes can no longer occur

Answer Key

1. B	2. A	3. C	4. D	5. A	6. D	7. C	8. B	9. C	10. D
11. A	12. C	13. B	14. C	15. B	16. B	17. C	18. C		

II

Short Answer Questions

Q.1 Define or describe very briefly the following:

(a) osmoregulation

Ans. Osmoregulation is the biological process by which organisms maintain a stable balance of water and electrolytes (salts) in their bodies, ensuring proper cellular and metabolic functions despite external environmental changes.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; significance – 1 mark

(b) Osmoregulators

Ans. Osmoregulators are organisms that actively control their internal osmotic concentration, maintaining homeostasis regardless of the osmolarity of their external environment. Examples include freshwater fishes and mammals.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark

(c) Osmoconformers

Ans. Osmoconformers are organisms that keep their internal osmotic pressure in equilibrium with their surrounding environment, without active regulation of solute concentrations. Many marine invertebrates are osmoconformers.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark

(d) Vasodilation

Ans. Vasodilation is the widening of blood vessels due to relaxation of smooth muscle in their walls. It increases blood flow to the skin and enhances heat loss, playing a role in thermoregulation.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; role in thermoregulation – 1 mark

(e) Vasoconstriction

Ans. Vasoconstriction is the narrowing of blood vessels caused by contraction of smooth muscle cells. It reduces blood flow to the skin, minimizes heat loss, and helps conserve core body temperature in cold environments.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; role in thermoregulation – 1 mark

(f) Thermoregulation

Ans. Thermoregulation is the ability of an organism to maintain its body temperature within a narrow range despite changes in external environmental conditions. It ensures proper enzyme activity and metabolic stability.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; significance – 1 mark

(g) Poikilotherms

Ans. Poikilotherms are animals whose internal body temperature fluctuates significantly with the external environment. Examples include amphibians, reptiles, and most fish.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark



(h) Homeotherms

Ans. Homeotherms are animals that maintain a relatively constant internal body temperature, independent of external environmental changes. Examples include birds and mammals.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark

(i) Ectotherms

Ans. Ectotherms are animals that depend mainly on external sources of heat, such as sunlight or warm surfaces, to regulate their body temperature. Examples include reptiles and amphibians.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark

(j) Endotherms

Ans. Endotherms are animals that produce most of their body heat through internal metabolic processes, enabling them to regulate temperature regardless of environmental fluctuations. Examples include birds and mammals.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; example / significance – 1 mark

(k) Homeostasis

Ans. Homeostasis is the dynamic process by which organisms maintain a stable internal environment—including temperature, pH, and fluid balance—despite external changes. It is vital for normal physiological functions and survival.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; significance – 1 mark

Q.2 What are the problems faced by osmoregulators?

Ans. Osmoregulators must maintain constant internal osmotic conditions despite fluctuations in external environments. In freshwater habitats, the problem is excessive water entry by osmosis and loss of salts. In contrast, in marine habitats, organisms face dehydration due to water loss and accumulation of excess salts. Both conditions disrupt homeostasis and require active physiological control. Osmoregulators use metabolic energy for active transport of ions, water reabsorption, and salt excretion through specialized organs such as kidneys, gills, or salt glands to maintain osmotic balance.

MARKING SCHEME (3 Marks)

- Problems in freshwater – 1 mark; problems in marine – 1 mark; energy cost/role of active transport – 1 mark

Q.3 Describe the osmoregulatory adaptations in (a) freshwater (b) marine (c) terrestrial animals.

- Ans.**
- (a) Freshwater animals:** Being hypertonic to their environment, they constantly gain water by osmosis. They excrete large amounts of dilute urine and actively absorb salts through gills or skin to replace lost ions.
 - (b) Marine animals:** Marine bony fishes are hypotonic to seawater, losing water by osmosis. They drink seawater and actively excrete excess salts via chloride cells in gills and kidneys. Marine cartilaginous fishes retain urea and TMAO, making their body fluids nearly isotonic to seawater.
 - (c) Terrestrial animals:** The main challenge is dehydration from evaporation and excretion. Adaptations include waterproof skin, behavioral changes such as nocturnality, kidneys producing concentrated urine, and efficient reabsorption of water in intestines and nephrons.

MARKING SCHEME (3 Marks)

- Freshwater adaptations – 1 mark; marine adaptations – 1 mark; terrestrial adaptations – 1 mark

Q.4 What are the needs of thermoregulation?

Ans. Thermoregulation is needed to maintain body temperature within a narrow physiological range. This ensures enzymes work at their optimal temperature, preventing denaturation or inactivation. Stable temperature also keeps metabolic rates constant, supporting efficient energy production and biochemical processes. Additionally, thermoregulation protects cellular structures and vital organs from damage due to extreme heat or cold. It is especially important for endotherms, enabling them to function in diverse and fluctuating environmental conditions.

MARKING SCHEME (3 Marks)

- Enzyme activity – 1 mark; metabolic stability – 1 mark; survival under extremes – 1 mark

Q.5 What is a feedback mechanism?

Ans. A feedback mechanism is a regulatory process in which the outcome of an action influences the system itself. In homeostasis, sensors detect

MARKING SCHEME (3 Marks)

- Definition – 1 mark; negative feedback – 1 mark; positive feedback – 1 mark



deviations from the set point and communicate with the control center, which triggers effectors to restore balance. Negative feedback counteracts the original change (e.g., regulation of body temperature or blood glucose), while positive feedback amplifies the response until a specific outcome is achieved (e.g., uterine contractions during childbirth). Feedback loops are vital for maintaining internal stability.

Q.6 Identify and define the four interacting components that maintain homeostasis in feedback loops.

Ans. Stimulus: Any detectable change in the internal or external environment that disrupts homeostasis, such as rising temperature or falling blood glucose.

MARKING SCHEME (3 Marks)
• Stimulus – 1 mark; receptor – 1 mark;
control center + effector – 1 mark

1. **Receptor (Sensor):** Detects the stimulus and sends information to the control center.
2. **Control Center:** Processes incoming information and determines the appropriate corrective response, often performed by the brain or endocrine glands.
3. **Effector:** Carries out the corrective action, such as muscles shivering to produce heat or glands secreting hormones to restore balance.

Q.7 Give two examples of physiological processes controlled by positive feedback loops.

Ans. Positive feedback amplifies changes until a specific process is completed. Two main examples include:

MARKING SCHEME (3 Marks)
• Childbirth example – 1.5 marks; blood clotting example – 1.5 marks

1. **Childbirth (parturition):** Stretching of the uterus stimulates the release of oxytocin. This hormone intensifies uterine contractions, which further increases oxytocin release. The loop continues until delivery is completed.
2. **Blood clotting:** Injury to a blood vessel triggers platelet adhesion at the site. Platelets release chemicals that attract more platelets, amplifying the clotting cascade until a stable clot seals the breach.

III

Long Answer Questions

Q.1 Classify animals on the basis of temperature.

Ans. Please See SLO [B-12-R-75], [B-12-R-76]

Q.2 What are the thermoregulatory strategies in man?

Ans. Please See SLO [B-12-R-77]

Q.3 How various human organs system to maintain controlled stable environment.

Ans. Please See [B-12-R-77]

Q.4 Explain components of homeostatic mechanism.

Ans. Please See SLO [B-12-R-78]

Q.5 Describe with example the negative feedback mechanism with relation to homeostasis.

Ans. Please See SLO [B-12-R-79]

Q.6 Describe with example the positive feedback mechanism with relation to homeostasis.

Ans. Please See SLO [B-12-R-79]

Q.7 Compare and contrast negative and positive feedback loops.

Ans. Please See [B-12-R-79]

Q.8 Explain how negative feedback controls body temperature.

Ans. Please See SLO [B-12-R-79]

